

Semantic Space Abstractions for Partially Observable Deterministic Task and Motion Planning

Final Presentation

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23rd Jun, 2026



Chair of Machine
Learning and Reasoning
(Computer Science 6)

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Roadmap

1. Introduction
2. Background
3. Related Work
4. Approach
5. Experimental Evaluation
6. Discussion

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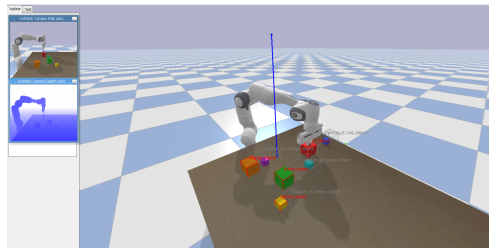
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- ▶ **But:** classical TAMP (such as PDDLStream) assumes a *fully observable, deterministic* world

Introduction: Planning under Occlusion

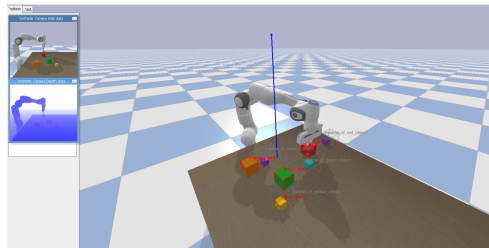
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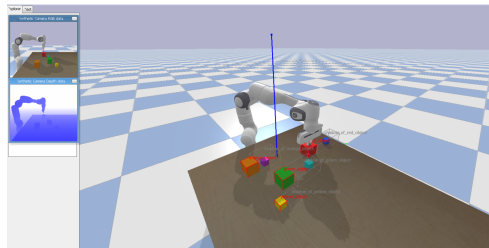
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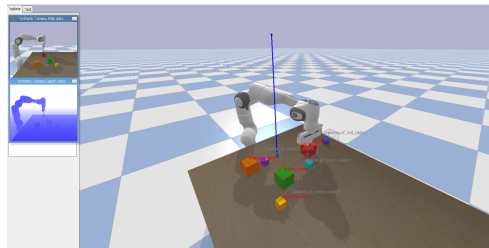
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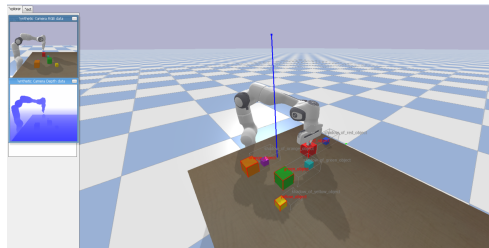
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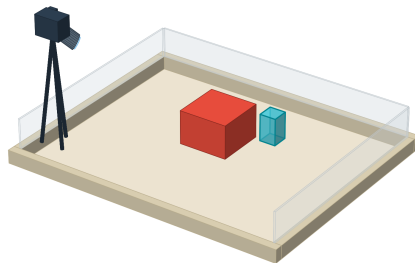


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Research question: How can we define and integrate a belief representation based on *semantic partitioning* into a partially observable deterministic (POD) TAMP framework?

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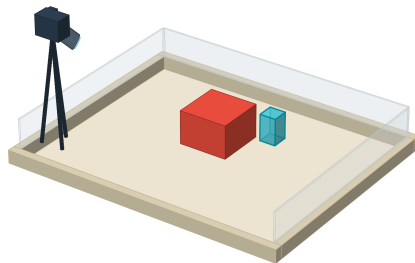
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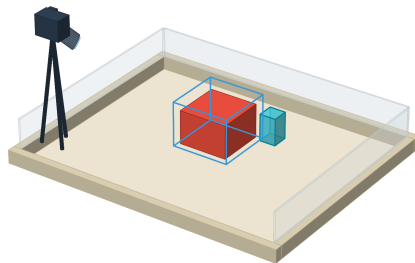
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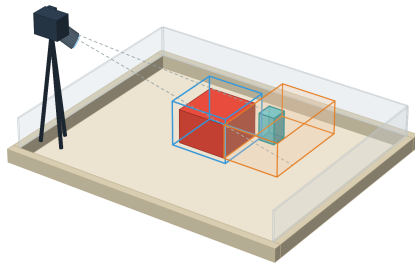
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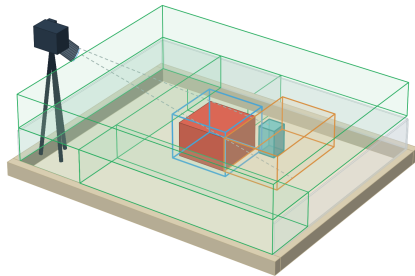
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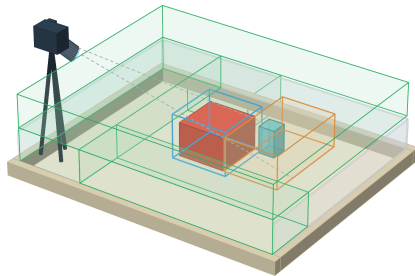
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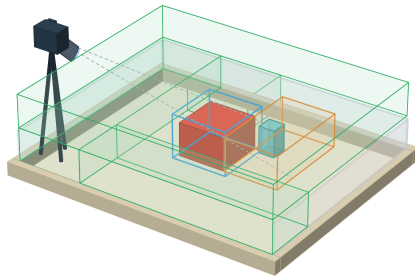
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 - ▶ covers free space
- ▶ Fine where it matters, coarse elsewhere — a handful of cells, not a dense grid
- ▶ A classical planner reasons over these few cells, and *looks* only where it must



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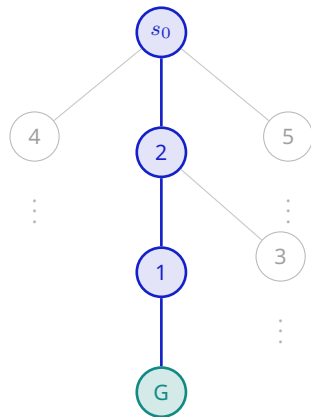
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- ▶ **STRIPS / PDDL** represents it compactly: a *factored* state is the set of true atoms, e.g. $\{(\text{on } A \ B), (\text{handempty})\}$; an action has preconditions and add / delete effects

Background: Solving It — Heuristic Search

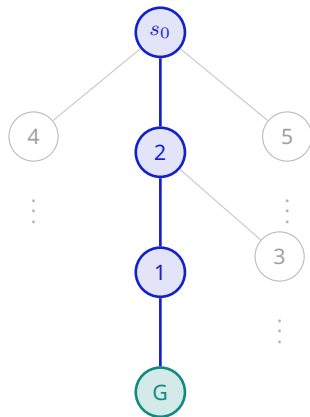
- Expand states **best-first**, ordered by a heuristic $h(s)$ — an estimate of the remaining distance to the goal



Labels: heuristic h . Lowest- h path in blue; grey branches are never expanded.

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- ▶ Expand states **best-first**, ordered by a *heuristic* $h(s)$ — an estimate of the remaining distance to the goal
- ▶ Promising states go first, so the goal is reached **without enumerating** the whole state space



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Background: When the World Is Not Fully Known

- ▶ Unlike classical planning, a robot rarely knows the **exact state**: it cannot see *behind* an object, so it cannot tell where a hidden object is — **partial observability**

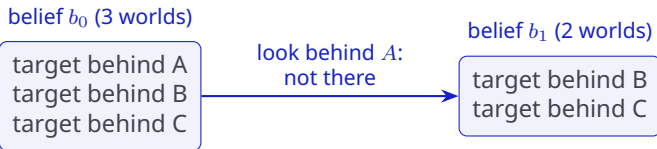
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Example — a target hidden behind one of three objects:



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Example — “is the cup in region B ?”

before sensing B

`obj_at_boxel(cup, B)`
= ? (don't read it yet)

`obj_at_boxel_KIF(cup, B)`
absent

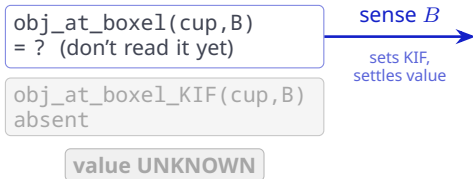
value UNKNOWN

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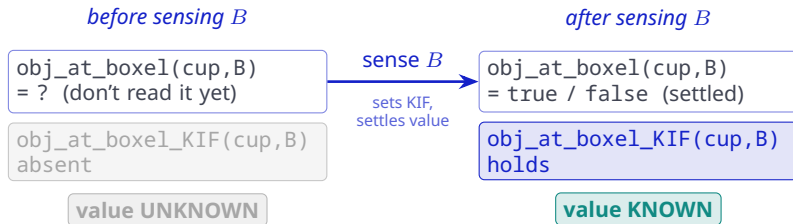
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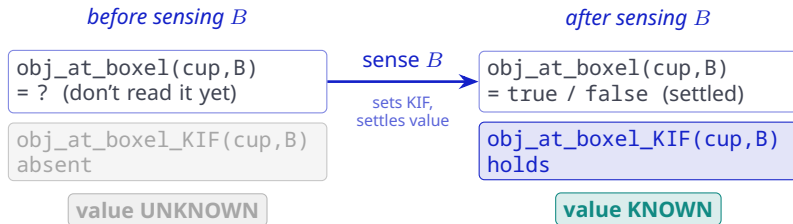
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This **compiles** partial observability into a **classical** problem: the planner now seeks a *state of knowledge* — the target's region becomes *known* (Geffner & Bonet [Gef+13]).

Background: What Kind of Uncertainty?

Two very different reasons a plan can be uncertain:

- ▶ **Random outcomes** — e.g. a move that occasionally slips sideways. This needs a **POMDP**: a probability distribution over states and a *policy* — expressive, but computationally very demanding (Kaelbling et al. [Kae+98])

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Ours is the second kind: the robot's actions are reliable — the only uncertainty is occlusion. So no probabilities; we track knowledge, and sensing resolves it.

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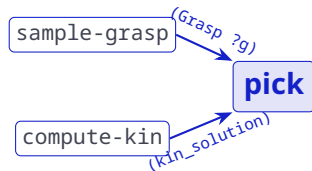
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Example — our pick action:

```
(:action pick :parameters (?o ?b ?g ?q)
:precondition (and (handempty)
  (obj_at_boxel ?o ?b)
  (at_config ?q)
  (Grasp ?g)
  (kin_solution ?o ?b ?g ?q)))
:effect (and (holding ?o) ...))
```

// from move
// sample-grasp



Each stream certifies one **blue** fact pick needs. The trajectory stream plan-motion feeds the preceding move, which sets **at_config**.

Related Work: TAMP under Partial Observability

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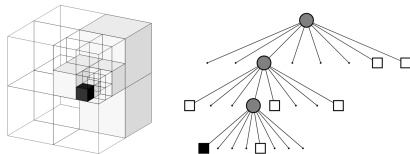
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- ▶ **Determinize-and-replan**: FF-Replan does this for probabilistic MDPs (Yoon et al. [Yoo+07]); the **K-replanner** does it for partial observability via knowledge literals (Bonet & Geffner [Bon+11]). We keep its replanning idea but *join it to TAMP* on named **Boxel** volumes — the K-replanner stays purely symbolic, with no geometry

Related Work: Spatial Representations under Occlusion

What we look for:

occlusion as *named, symbolic state* a planner can resolve.

- **Octree maps** (OctoMap): map *where matter is*; unobserved space is only absent evidence, and subdivision is geometric, not semantic (Hornung et al. [Hor+13])



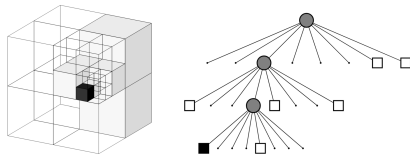
An octree map: free (white) and occupied (black) cells, with its tree (after Hornung et al.).

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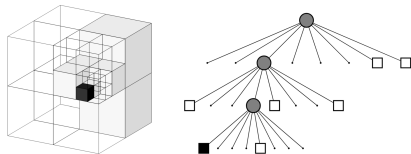
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- ▶ **Learned critical regions**: abstract space by *task relevance*, but predicted once per scene and carry no occlusion (Molina et al. [Mol+20])
- ▶ **Occlusion-aware planners**: keep occlusion in a *probabilistic belief* (visibility grid or pose distribution), not a named volume (Garrett et al. [Gar+20])



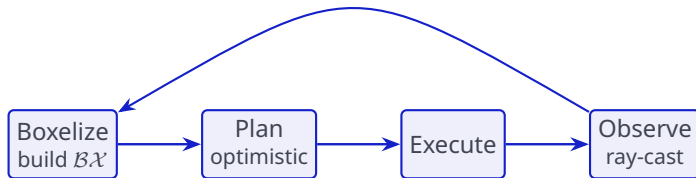
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Approach: The Sense-Plan-Act Loop

The method, end to end:

boxelize the scene, plan optimistically, execute, observe — then loop. The rest of this section fills in each step.

re-boxelize and replan if reality disagrees



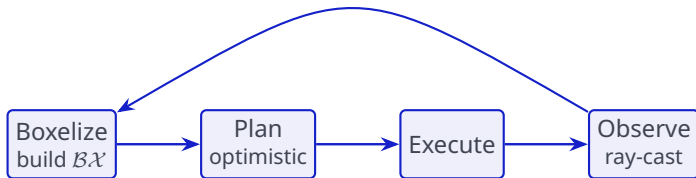
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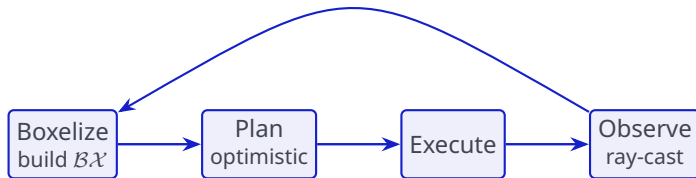
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- ▶ Plan optimistically, execute, **repair by replanning**
- ▶ **Re-boxelize** when a placement consumes a cell or a new occluder appears
- ▶ **Time-bounded**: ends on goal, no plan, search exhausted, or budget

Approach: Building Boxels

Start from the objects, the camera, and the workspace — no Boxels yet

1. Object Boxels

wrap each object in a tight box

2. Shadow Boxels

add the region each object hides from the camera

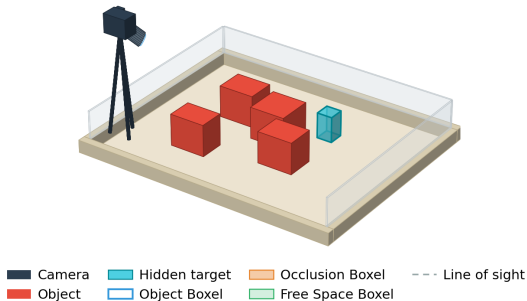
3. Free-space Boxels

begin with the whole workspace as one cell
a cell that is empty becomes a free Boxel;
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4. Merge

free Boxels that share a face

(a) Detected scene



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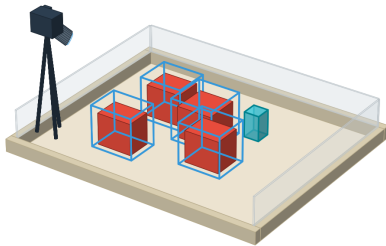
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(b) Object Boxels



■ Camera	■ Hidden target	■ Occlusion Boxel	--- Line of sight
■ Object	■ Object Boxel	■ Free Space Boxel	

Approach: Building Boxels

Start from the objects, the camera, and the workspace — no Boxels yet

1. Object Boxels

wrap each object in a tight box

2. Shadow Boxels

add the region each object hides from the camera

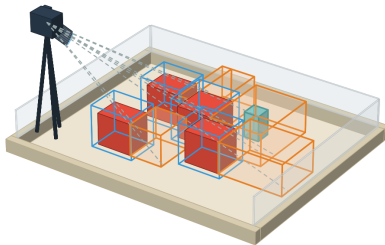
3. Free-space Boxels

begin with the whole workspace as one cell
a cell that is empty becomes a free Boxel;
otherwise cut it into 8 smaller cells,
stopping at a minimum size

4. Merge

free Boxels that share a face

(c) Occlusion Boxels



■ Camera ■ Hidden target ■ Occlusion Boxel - - - Line of sight
■ Object ■ Object Boxel ■ Free Space Boxel

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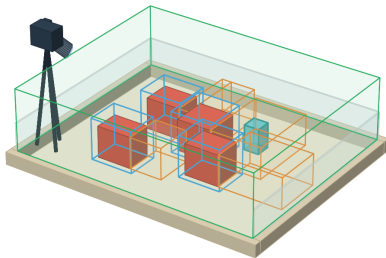
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(d) Free space: one cell



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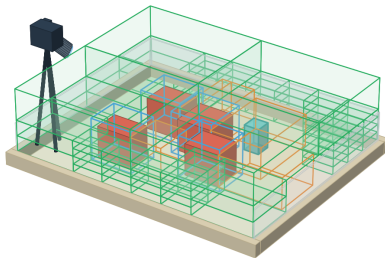
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(e) Octree split



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Object	Object Boxel	Free Space Boxel	

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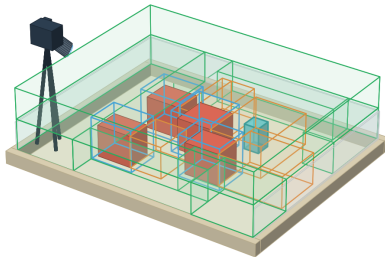
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(f) Convex merge



■ Camera	■ Hidden target	■ Occlusion Boxel	--- Line of sight
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Approach: Belief over Boxels

- ▶ The Boxel partition *is* the belief support: a hidden target gives **one candidate world per Boxel** it could occupy ($\mathcal{S}_0$)

Example: target t behind one of 3 occluders

belief $\mathcal{S}_0$: 3 worlds

KIF(t, b_1)	=	?
KIF(t, b_2)	=	?
KIF(t, b_3)	=	?

$\text{KIF}(t, b_i)$: is target t in shadow Boxel b_i ? The unknown (?) ones are the candidate worlds.

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belief: 2 worlds

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- ▶ The Boxel partition *is* the belief support: a hidden target gives **one candidate world per Boxel** it could occupy ($\mathcal{S}_0$)
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- ▶ Carried by **Know-If fluents**, so a plain classical planner reasons over the hidden object — **no separate belief solver**

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Approach: PDDLStream Realization

The sensing action (optimistic, calls no streams)

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sense(?o, ?region)
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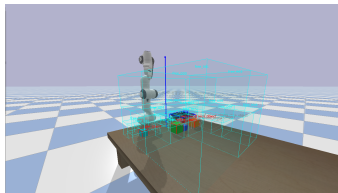
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 - ▶ plan-motion — a collision-free trajectory (RRT-Connect (Kuffner & LaValle [Kuf+00]))

Experimental Evaluation: Questions & Setup

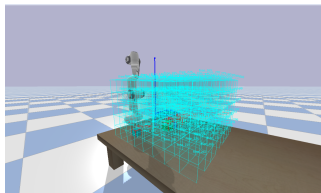
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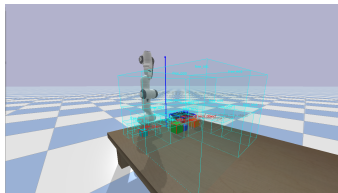
semantic



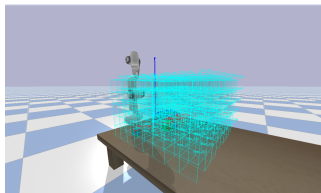
uniform

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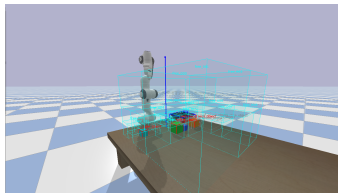


uniform

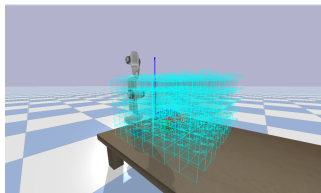
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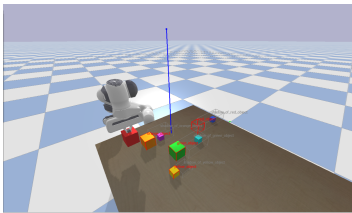


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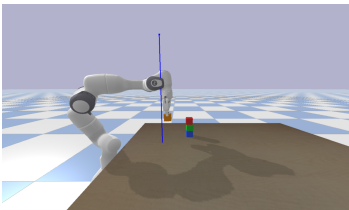
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Setup: PyBullet + 7-DOF Franka Panda, one fixed camera, **oracle** perception; 3 goals $\times$ 3 difficulty levels $\times$ 30 seeds, 1800 s budget each. Oracle perception isolates the representation — an ablation, not a benchmark win.

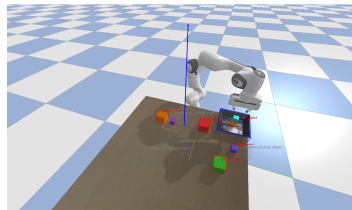
Experimental Evaluation: Tasks



holding: retrieve a hidden target

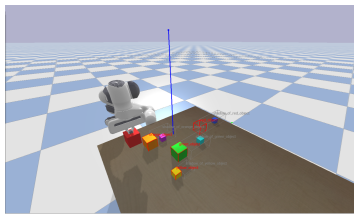


stack: build a tower to height h

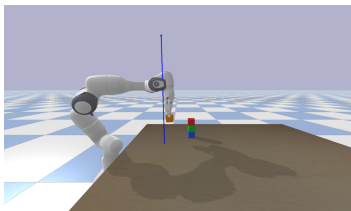


find-and-tray-stack: find, then stack on a tray

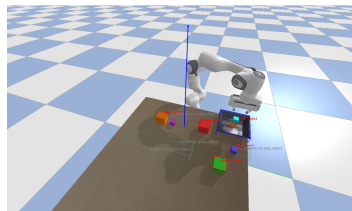
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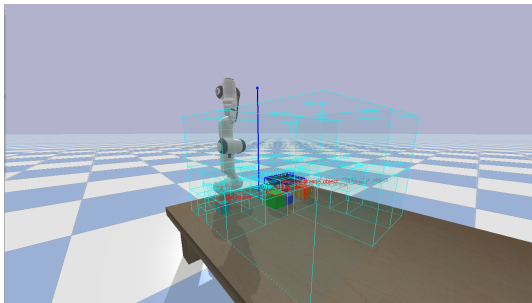
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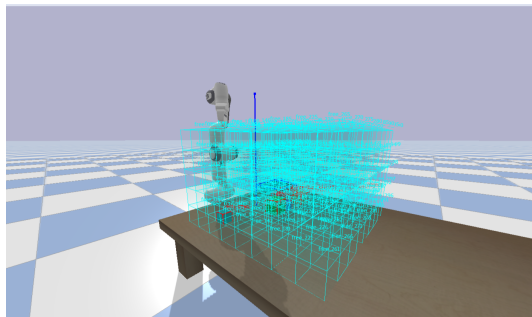
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- Difficulty axis: number of occluders $n_{\text{occ}} \in \{2, 3, 4\}$ (search goals), or stack height $h \in \{2, 3, 4\}$ — each adds shadow regions or tower layers

Results: Semantic vs Uniform Partition



semantic: $\approx 22\text{--}38$ task-relevant Boxels



uniform: ~ 300 fixed cells over the whole workspace

Same cell size, same planner — the adaptive partition grounds and searches an order of magnitude fewer cells.

Results: Success and Planning Time

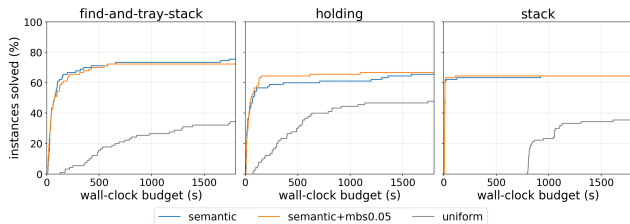
goal	variant	success	mean plan time (s)
find-and-tray-stack	semantic	75.6 %	124.6
	uniform	34.4 %	692.1
holding	semantic	65.6 %	134.3
	uniform	47.8 %	438.5
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Same planner, same cell size: the adaptive partition solves $\approx 1.4\text{--}2.2\times$ more scenes, while planning several-fold to $\sim 50\times$ faster.

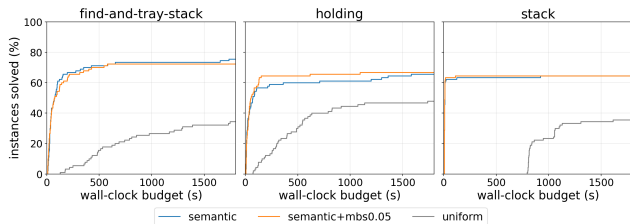
Results: The Cost Gap



- The adaptive partition **dominates at every budget**; uniform starts later and plateaus lower

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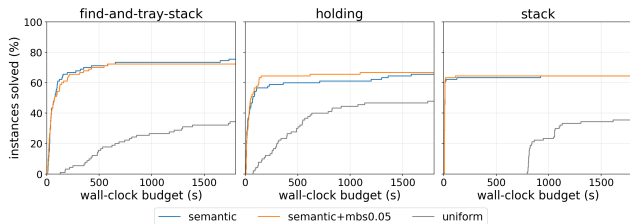
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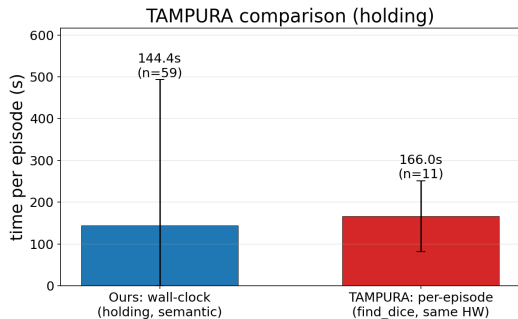
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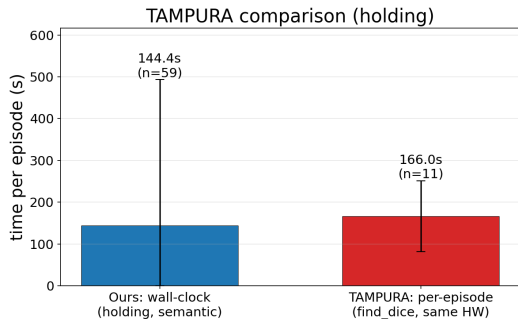
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Results: Comparison with TAMPURA



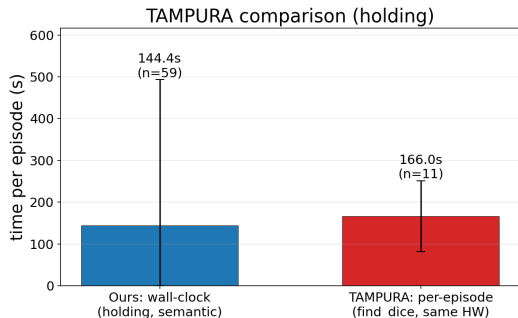
- Rough **parity** end-to-end (≈ 144 s vs 166 s per episode) and comparable reliability (65.6 % vs the ≥ 63 % its returns imply)

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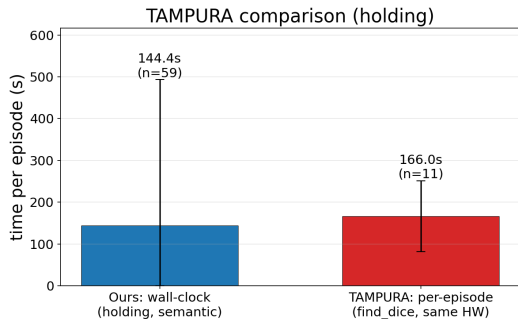
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- ▶ We hold occlusion as **symbolic state** — line of sight is a planning condition; TAMPURA keeps it sub-symbolic
- ▶ Not a benchmark win: different environments, same hardware

Discussion: Limitations

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- ▶ **Box-shaped shadows under-cover the hidden region:** the volume an object occludes is really a cone, so a target tucked into the uncovered part gets no Boxel to sense — our most common failure
- ▶ **Bounded give-up:** on a permanently blocked shadow the planner eventually stops searching — unsound, since the target may still be reachable (we determinize-and-replan, not full contingent POD)

Discussion: Future Work & Takeaway

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a real-robot Franka study

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-  a full semantic free-space merge

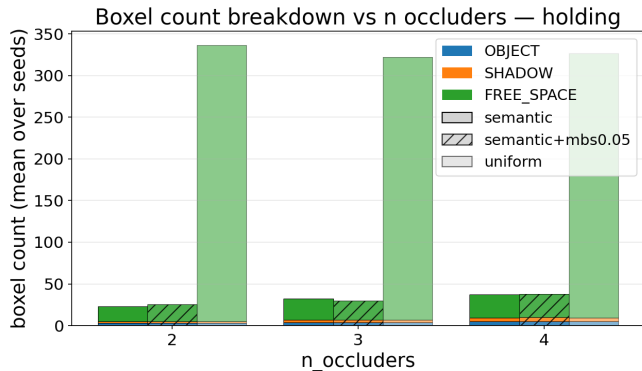
Takeaway: making occlusion an explicit, named piece of symbolic state lets a simple *deterministic* planner do partially observable TAMP — reasoning about where to look — without learning a probabilistic policy.

Summary

- ▶ **Problem:** TAMP under occlusion — the robot must decide *where to look*, but a dense voxel belief is too large to plan over
- ▶ **Idea: Boxels** — task-relevant box-shaped cells (object / shadow / free) that put the structure of space into the symbolic state
- ▶ **Method:** a deterministic POD-TAMP planner reasons over Boxels with Know-If fluents; optimistic sensing + replanning resolve the unknown
- ▶ **Result:** an order of magnitude fewer cells $\Rightarrow$ far more cluttered scenes solved than a uniform grid; at parity with TAMPURA, without a learned probabilistic policy

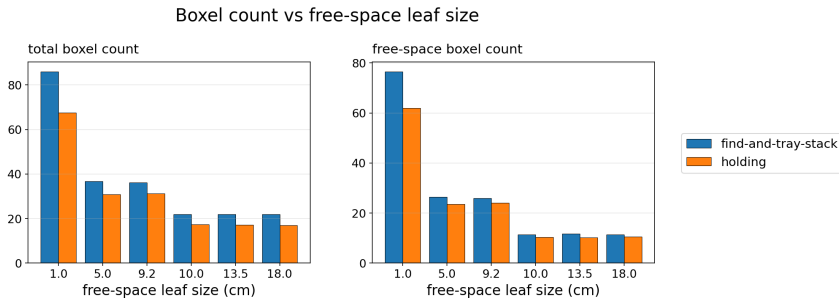
Thank you! Questions?

Backup: Representation Compactness



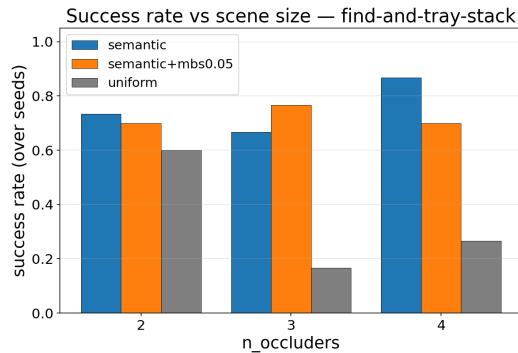
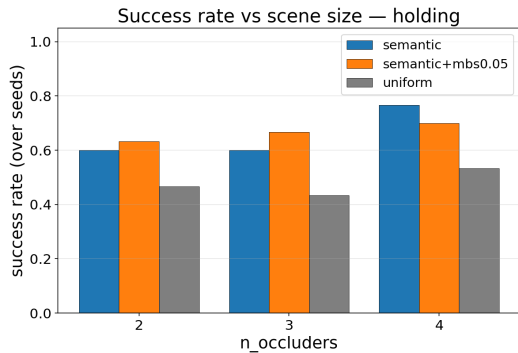
- ▶ Adaptive: $\approx 22-38$ Boxels in total
- ▶ uniform: $\sim 300+$ free-space cells alone
- ▶ An order of magnitude fewer cells to ground and search

Backup: Free-Space Resolution



The partition is **insensitive** to the free-space leaf size: flat from 5 cm to the auto setting; only the extreme 1 mm floor breaks feasibility (too fine to plan within budget).

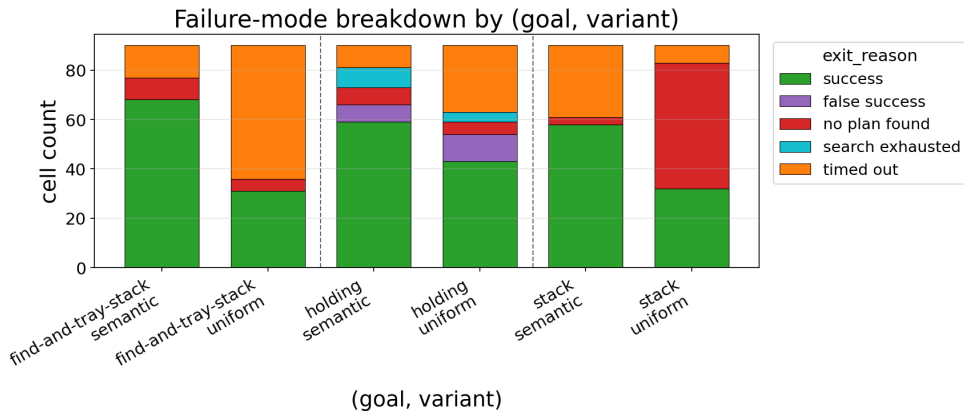
Backup: Scaling with Clutter



More occluders do not break the adaptive partition: it stays well above uniform at every clutter level, and even rises at the densest scenes.

Orange is `semantic+mbs0.05` — the same partition with a 5 cm floor on free-cell size (a resolution check); it tracks `semantic.stack` instead scales with tower *height*, not clutter — a manipulation limit, not the planner.

Backup: Failure Modes



Exit reason per (goal, variant), 90 runs each. **uniform** fails mostly by *timing out* (and *no plan found* on stack): it cannot ground and search ~ 300 cells within budget. **semantic**'s residual failures are fewer, split across *timed out*, *search exhausted* (target outside every box-shaped shadow), and *false success* (a physics mismatch at execution).